

TEMPEST: Porewater SO₄/Cl

202606 81-151 Samples

2026-09-02

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##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Sample_Year = "2026"
Date_Run = "2026-09-02" #Date that instrument was run
Run_by = "Zoe Read" #Instrument user
Script_run_by = "Zoe Read" #Code user
run_notes = "
" #any notes from the run

##Sample data that was entered incorrectly
# The Old ID is the original, incorrectly-entered ID and the New ID is the correct ID to change it to.
Old_ID_1 = NA
New_ID_1 = NA

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/COMPASS_TEMPEST_202606_82-151_C1.txt"
raw_file_name_so4 = "Raw Data/COMPASS_TEMPEST_202606_82-151_S04.txt"

#Sample ID Path
SampleID_path = "Raw Data/TEMPEST_S04_20260501-20260612_corrected.csv"

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_TEMPEST_Processed_C1_S04_202606_82-151.csv"

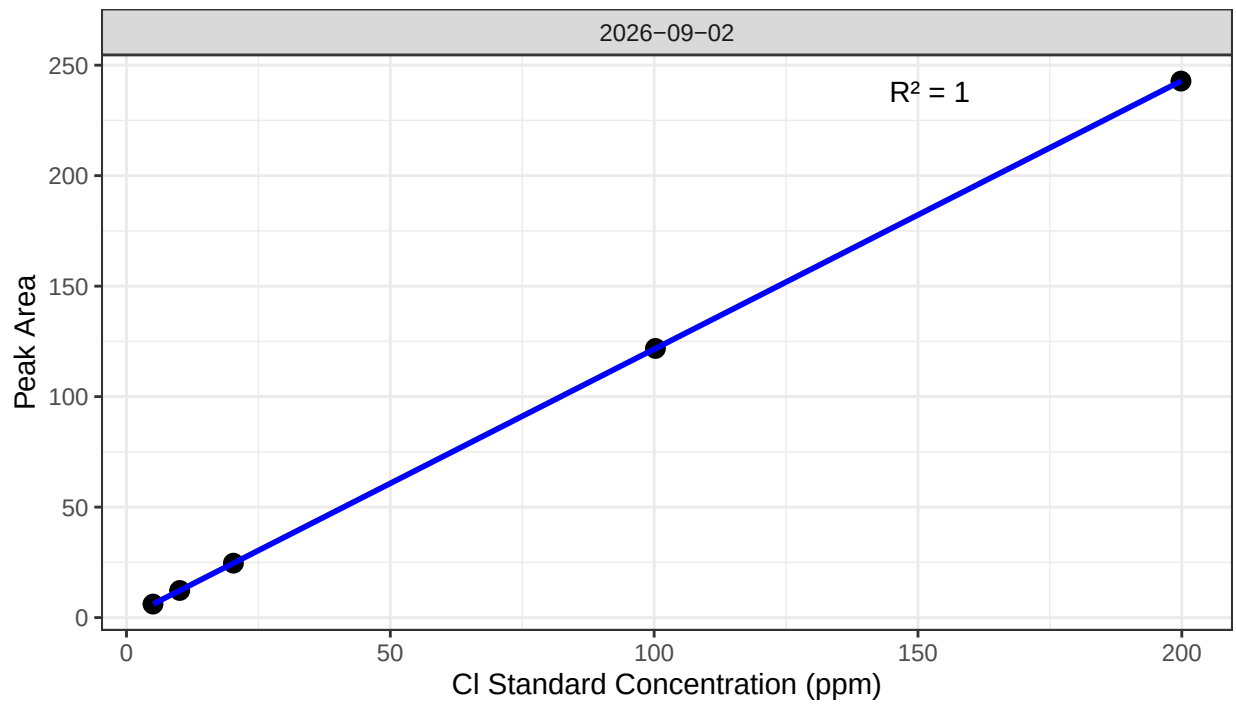
##### Log Files - PLEASE CHECK

#qaqc log file path for this year copied over from COMPASS GitHub
Log_path = "Raw Data/COMPASS_Synoptic_C1_S04_QAQClog_2026.csv"

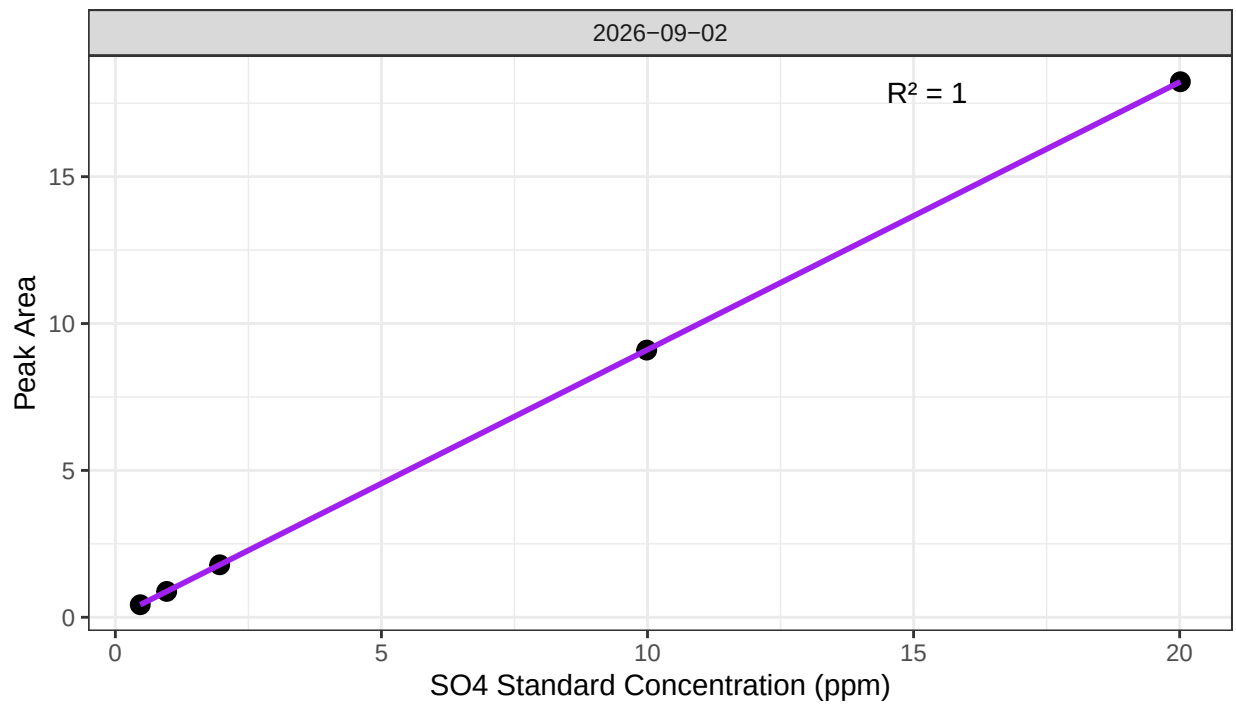
##Set Up Code - constants and QAQC cutoffs
##Import Sample Data
##Fix Sample IDs entered wrong
```

0.2 Assess Standard Curves

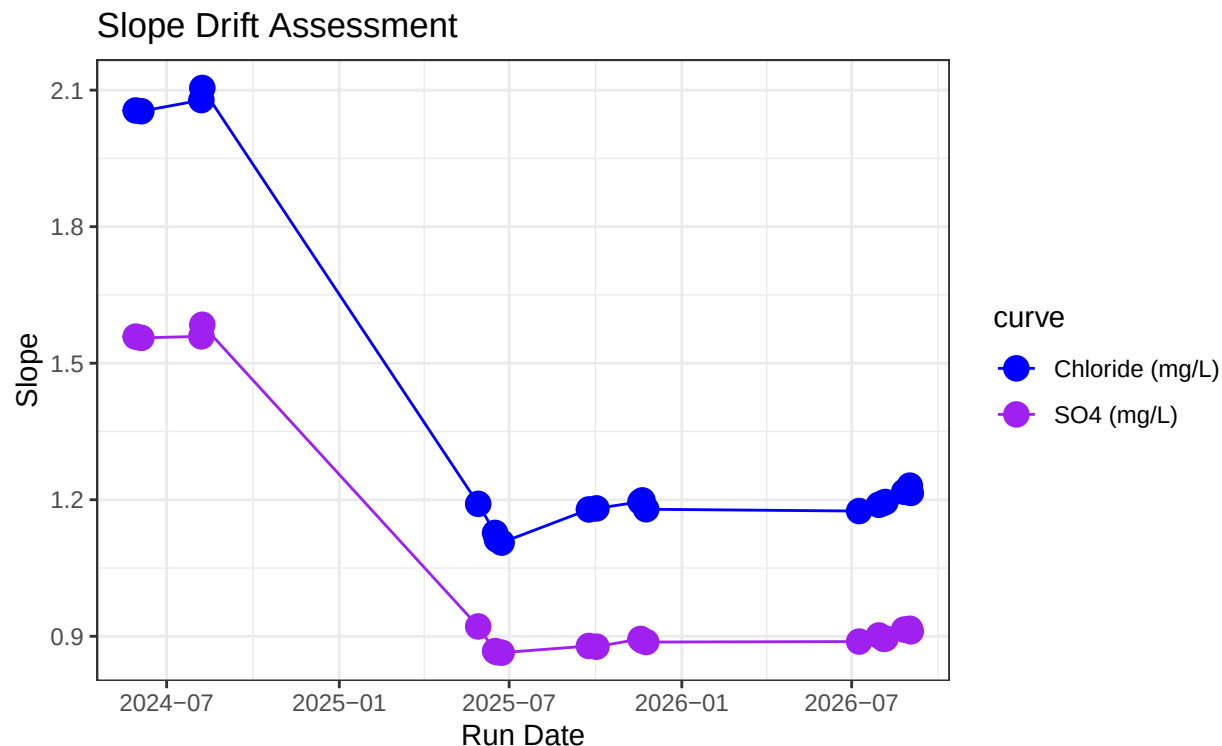
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

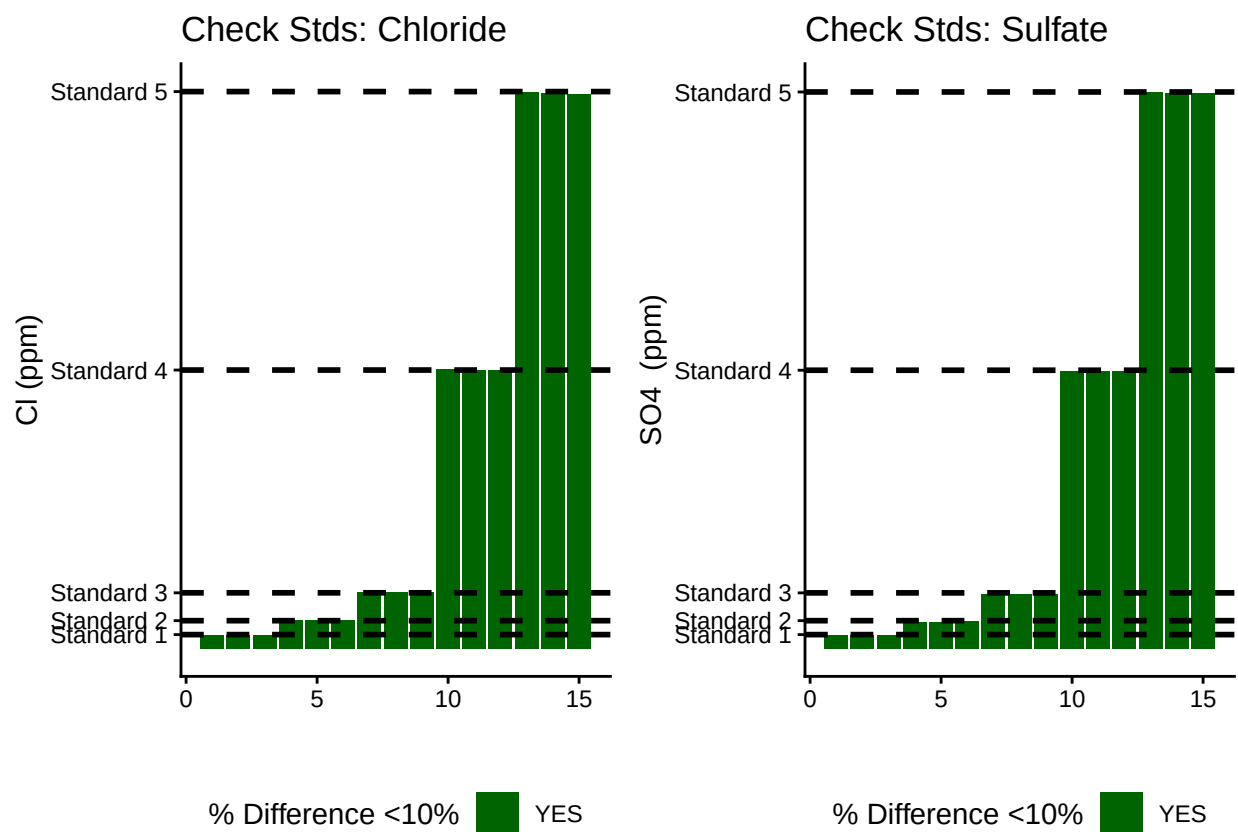
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl   sd_Cl   cv_Cl flag_Cl
##   <chr>      <dbl>   <dbl>   <dbl> <chr>
## 1 Standard 1    5.03 0.000950 0.000189 Chloride Check Standard RSD within Range~
## 2 Standard 2   10.1 0.00394 0.000391 Chloride Check Standard RSD within Range~
## 3 Standard 3   20.3 0.0155 0.000765 Chloride Check Standard RSD within Range~
## 4 Standard 4  100. 0.109 0.00109 Chloride Check Standard RSD within Range~
## 5 Standard 5  200. 0.288 0.00144 Chloride Check Standard RSD within Range~
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4 cv_SO4 flag_SO4
##   <chr>      <dbl>   <dbl>   <dbl> <chr>
## 1 Standard 1    0.476 0.00726 0.0152 Sulfate Check Standard RSD within Range --
## 2 Standard 2    0.969 0.00489 0.00504 Sulfate Check Standard RSD within Range --
## 3 Standard 3    1.97 0.00516 0.00263 Sulfate Check Standard RSD within Range --
## 4 Standard 4    9.97 0.0140 0.00141 Sulfate Check Standard RSD within Range --
## 5 Standard 5   20.0 0.0364 0.00182 Sulfate Check Standard RSD within Range --
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



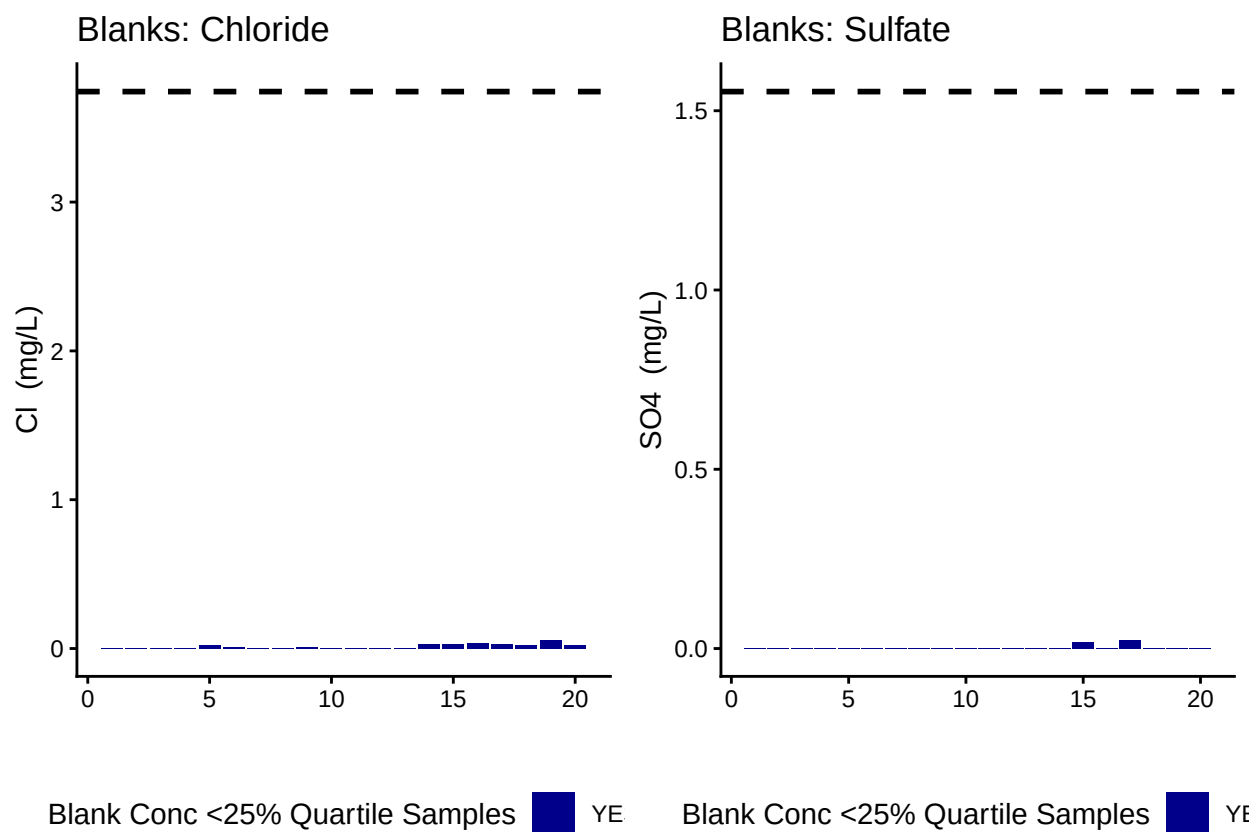
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



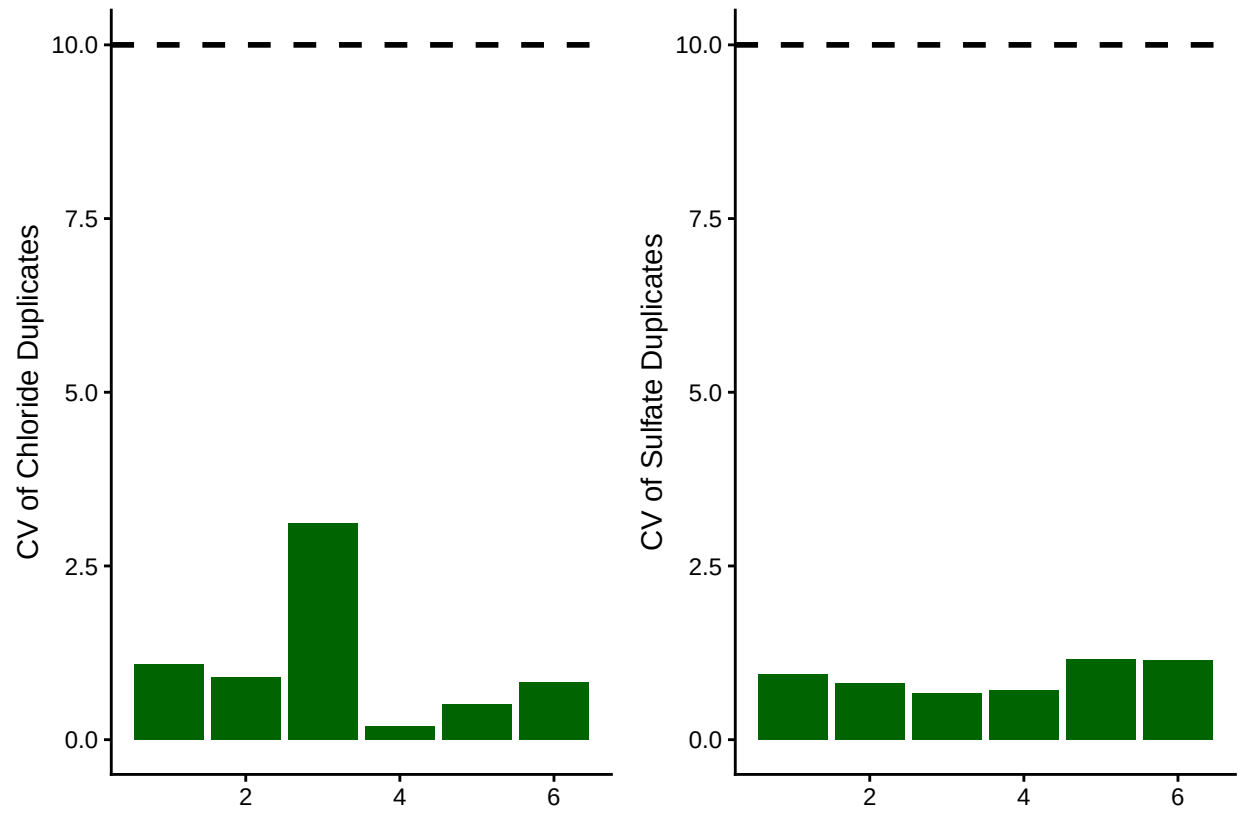
```
## Chloride blanks mean ppm:
```

```
## [1] 0.013605
```

```
## Sulfate blanks mean ppm:
```

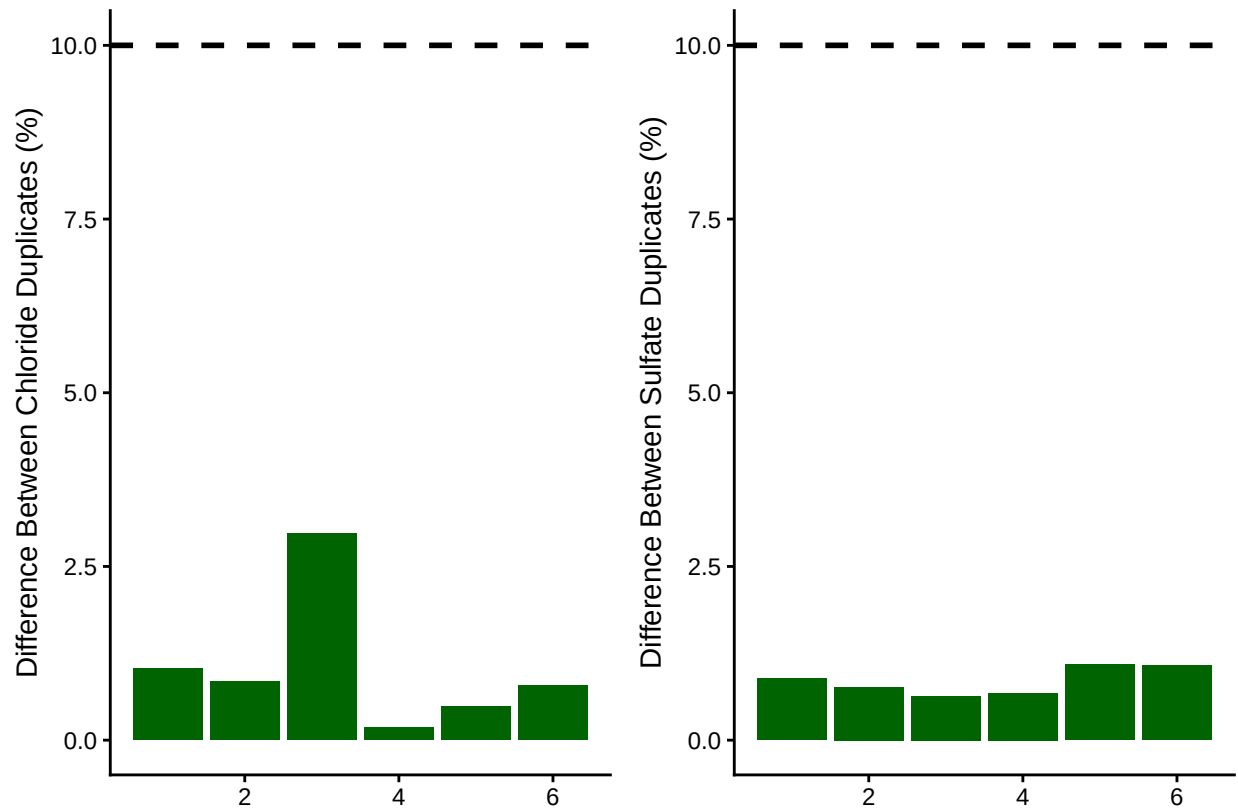
```
## [1] 0.00223
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$SO4_Conc_mM <- (all_dat$SO4_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for TEMPEST this depends on the sample so...
all_dat$Dilution <- 50
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "TMP"), 50, all_dat$Dilution)
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "EST_SourceWater"), 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "SW_SourceWater"), 100, all_dat$Dilution)
```



```

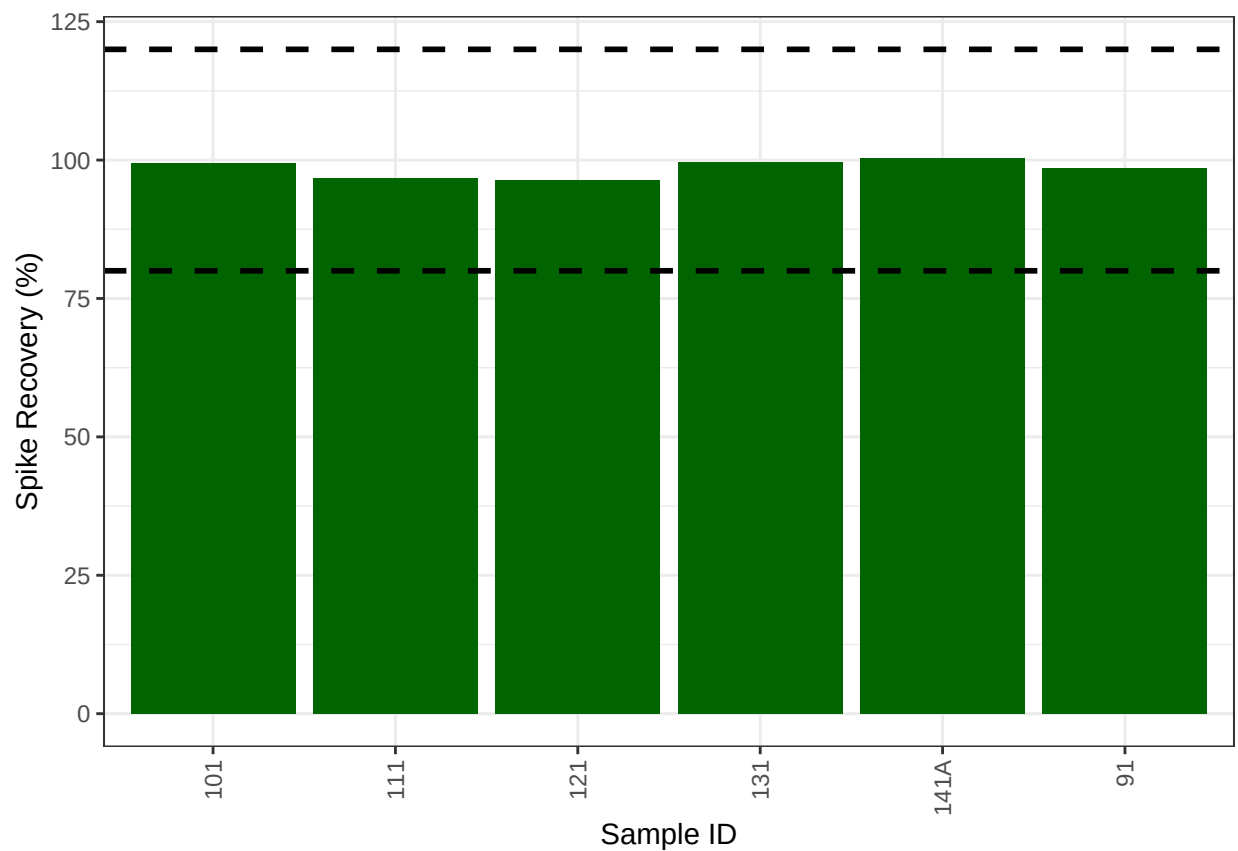
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "FATE"), 100, all_dat$Dilution)

all_dat$Dilution <- ifelse(all_dat$sample_ID == "90", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "92", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "93", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "94", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "129", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "132", 100, all_dat$Dilution)

# head(all_dat)

```

0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

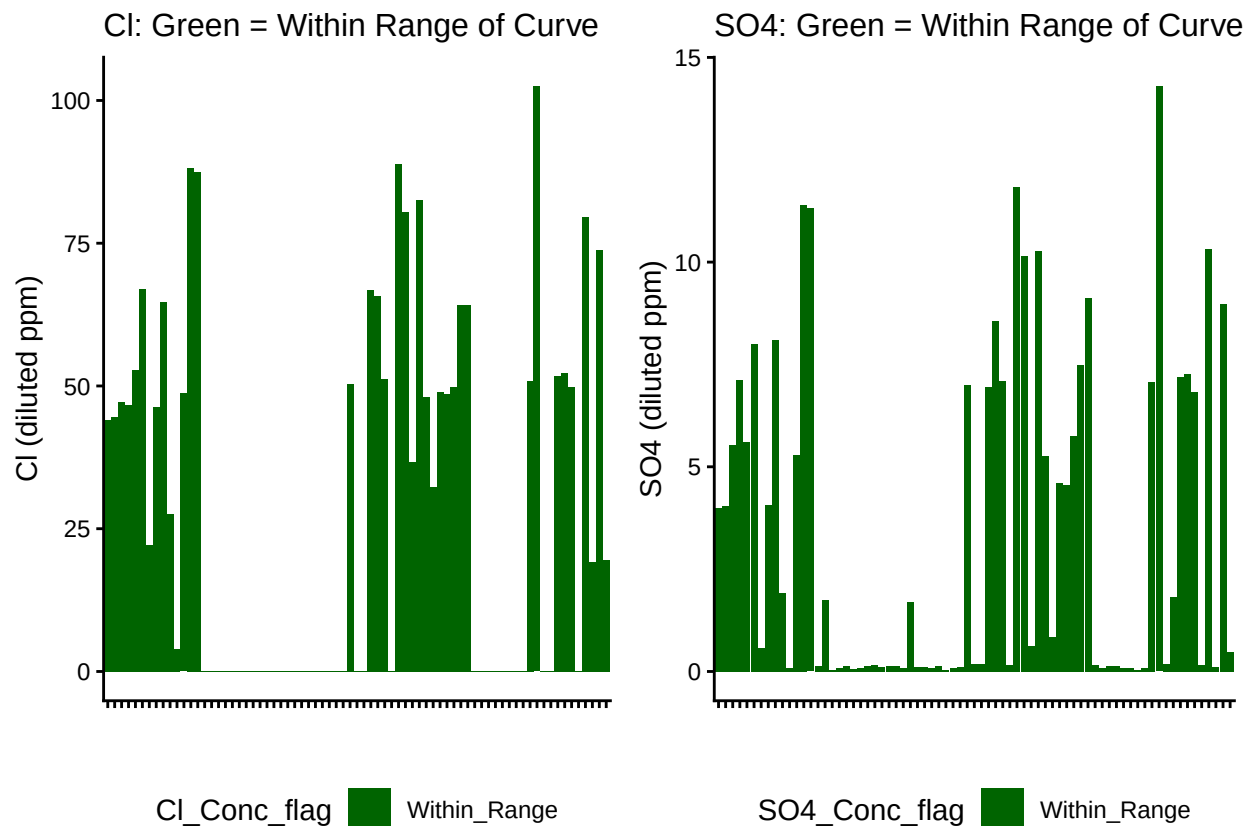


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

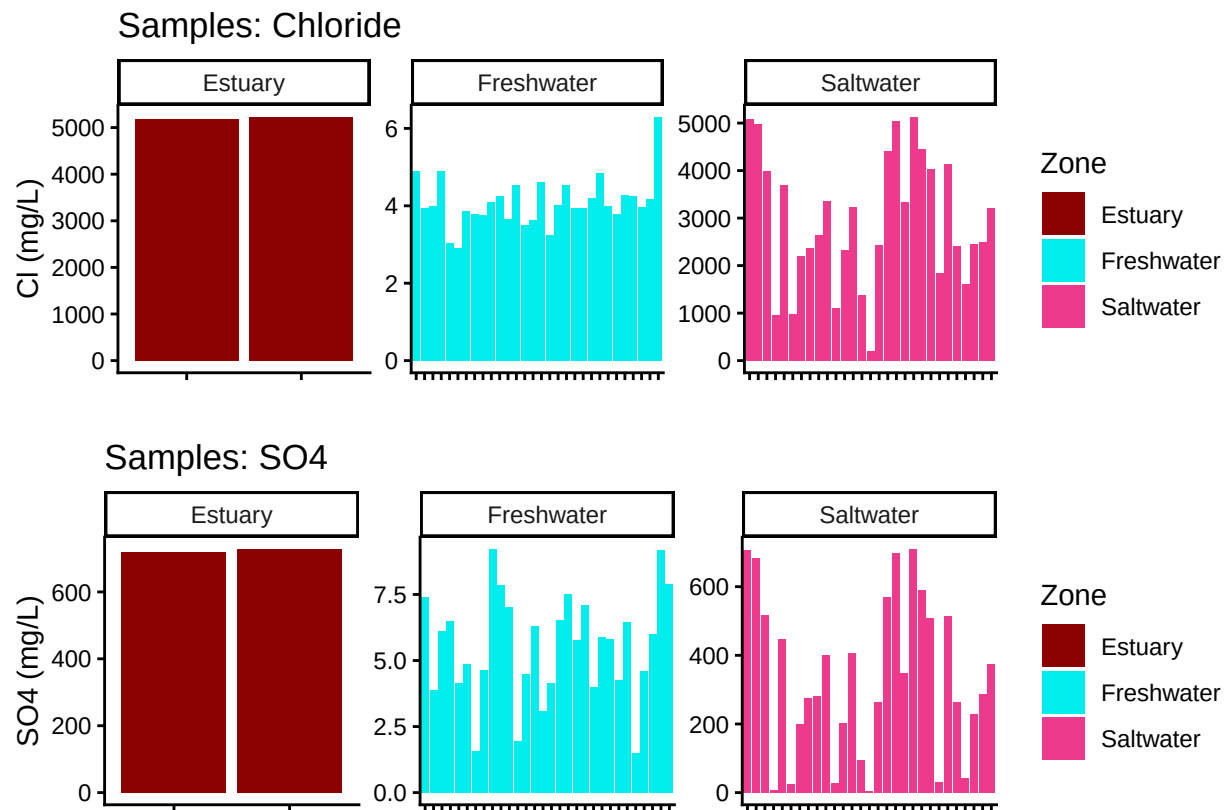
Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

```
##Merge with Sample ID list to get actual sample IDs
##Pull in active porewater inventory
##Check samples against metadata

## ***All sample IDs are present in metadata.***
```

0.9 Visualize Data by Plot



0.10 Export Processed Data

#end